

Labor Exploitation

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Abstract

In an employee-employer relationship, an employee typically cannot fulfill all of his or her biological needs, such as housing and healthcare, without working. This provides an advantage to the employer in setting the price of employment. Since biological needs are highly price- and income-inelastic, an employer can reduce the price of employment by negotiating a lower wage, requiring longer or worse working hours, or requiring work to be done under worse conditions than the employee would otherwise accept if his or her biological needs were more accessible. The resulting change in the slope of the demand curve for these needs results in value extracted by the employer from the employee due solely to the employee's inability to acquire his or her biological needs without working. The rapidly intensifying exploitation of young workers by predominately older corporate shareholders via this mechanism—in addition to a high tax burden on the young to subsidize predominately older government beneficiaries while the young themselves are faced with a deteriorating safety net and declining labor protections—has resulted in a *decreased* real median income per unit output for young male workers, a severe decline in fertility in “developed” and “developing” countries, and a decline in youth physical and mental health in the United States. Solutions to these issues are highly cost-effective but must be implemented rapidly to avoid severe declines in youth health, catastrophic population loss, and collapse of social institutions.

Elasticity

Elasticity is a measure of the change in one variable in response to a change in another, taking into account the initial values of each variable. Specifically, it is the ratio of the percentage change in one variable to a percentage change in another. For a discrete function, the x elasticity of y is defined as:

$$\epsilon_{x,y} = \frac{\frac{y_1 - y_0}{y_0}}{\frac{x_1 - x_0}{x_0}}$$

$$= \frac{\frac{\Delta y}{y_0}}{\frac{\Delta x}{x_0}}$$

$$= \frac{x_0}{y_0} \frac{\Delta y}{\Delta x}$$

where:

x_0 and x_1 are the initial and final values of x , respectively, and

y_0 and y_1 are the initial and final values of y , respectively.

For a continuous function, the x elasticity of y at (x_0, y_0) is defined as:

$$\epsilon_{x,y} = \lim_{\Delta x \rightarrow 0} \left(\frac{x_0}{y_0} \frac{\Delta y}{\Delta x} \right)$$

$$= \frac{x_0}{y_0} \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$$= \frac{x_0}{y_0} \frac{dy}{dx}$$

Price Elasticity of Demand

The **price elasticity of demand** is a measure of the change in demand for a good or service in response to a change in price, taking into account the initial demand for the good or service (q_0) and the initial price (p_0). Specifically, it is the ratio of the percentage change in demand for a good or service to a percentage change in price. For a discrete demand function, the price elasticity of demand is defined as:

$$\epsilon_{p,q} = \frac{p_0}{q_0} \frac{\Delta q}{\Delta p}$$

where:

q is the demand for the good or service and

p is the price of the good or service.

For a continuous demand function, the price elasticity of demand is defined as:

$$\epsilon_{p,q} = \frac{p_0}{q_0} \frac{dq}{dp}$$

Income Elasticity of Demand

The **income elasticity of demand** is a measure of the change in demand for a good or service in response to a change in income, taking into account the initial demand for the good or service (q_0) and the initial income (m_0). Specifically, it is the ratio of the percentage change in demand for a good or service to a percentage change in income. For a discrete demand function, the income elasticity of demand is defined as:

$$\epsilon_{m,q} = \frac{m_0}{q_0} \frac{\Delta q}{\Delta m}$$

where:

q is the demand for the good or service and

m is income.

For a continuous demand function, the income elasticity of demand is defined as:

$$\epsilon_{m,q} = \frac{m_0}{q_0} \frac{dq}{dm}$$

Elastic, Unit Elastic, and Inelastic Demand

For $|\epsilon| < 1$, demand is said to be **inelastic**; for $|\epsilon| = 1$, demand is said to be **unit elastic**; and for $|\epsilon| > 1$, demand is said to be **elastic**. Demand is said to be **perfectly inelastic** if $\epsilon = 0$ and **perfectly elastic** if $\epsilon \rightarrow -\infty$ or $\epsilon \rightarrow \infty$. This is summarized in the table below:

ϵ	Label
$ \epsilon < 1$	Inelastic
$ \epsilon = 1$	Unit elastic
$ \epsilon > 1$	Elastic
$\epsilon = 0$	Perfectly inelastic
$ \epsilon \rightarrow \infty$	Perfectly elastic

Net Benefit and Opportunity Cost

Net Benefit

Let $f(x)$ be an individual's **net benefit function** that takes as input an action x and produces as output the net benefit—or value—of that action for the individual, accounting for the cost, if any, of choosing that action. The net benefit function is defined as:

$$f(x) = b_x - c_x$$

where:

b_x is the benefit obtained from choosing action x (without accounting for the cost of choosing action x), and

c_x is the cost incurred from choosing action x .

Opportunity Cost

Suppose that an individual is faced with choosing one of two actions, action A or action B. Then, the **opportunity cost** of choosing action B instead of action A is $f(A) = b_A - c_A$ —that is, the net benefit of choosing action A that is given up when choosing B instead. Thus, when

an individual is faced with choosing to purchase one of two goods, good A or good B, the opportunity cost of purchasing good B instead of good A is:

$$f(A) = b_A - p_A$$

where:

b_A is the benefit (value) to the individual from choosing good A, and

p_A is the cost (price) of good A.

The benefit b_x of choosing action X in the individual's net benefit function will fluctuate in accordance with increasing or decreasing quantities of goods that an individual is faced with purchasing. Suppose an individual is faced with choosing between purchasing a combination (x, y) of two common consumer goods, good A and good B, with x being the amount of good A the individual will purchase and y the amount of good B. For an individual to accept a very small quantity of A—that is, very small x —the individual will typically require a disproportionately larger quantity of B—that is, very large y —to achieve the same utility (value), such that $\Delta y > \Delta x$. That is, as $x \rightarrow 0$, $y \rightarrow \infty$.

Stone-Geary Utility and Utility Maximization Subject to the Budget Constraint

Stone-Geary Utility

An individual's **utility function** $U(x_1, \dots, x_n)$ takes as input n quantities $x_1, \dots, x_n$ of n goods, respectively, and outputs a **utility**—that is, a value—to the individual of possessing those goods. The **Stone-Geary utility function** is a standard utility function that accounts for minimum levels of consumption of goods due to subsistence (that is, the minimum level of goods that the individual must consume in order to survive). It has the form:

$$U(x_1, \dots, x_n) = \prod_{j=1}^n (x_j - z_j)^{\alpha_j}$$

where:

$z_j \geq 0$ is the minimum quantity of good x_j demanded by the individual for subsistence, and

a_j is a nonnegative weight such that $\sum_{i=1}^n a_j = 1$.

Since Stone-Geary utility functions are well-suited to model the utility of goods that require a certain quantity for subsistence (such as water or food, for example), in this paper we will deal only with Stone-Geary utility functions. For simplicity, we will deal only with Stone-Geary utility functions of two variables x and y representing the quantities of two goods A and B, respectively:

$$U(x, y) = (x - z_A)^\alpha (y - z_B)^{1-\alpha}$$

In the Stone-Geary utility function, α and $1 - \alpha$ indicate preference for goods A and B, respectively, with increasing values indicating increasing preference and decreasing values indicating decreasing preference. Specifically, as will be shown later, α is the proportion of leftover income that will be spent on good A after all subsistence expenditures have been met:

$$p_A x = p_A z_A + \alpha(m - p_A z_A - p_B z_B)$$

Similarly, $1 - \alpha$ is the proportion of leftover income that will be spent on good B after all subsistence expenditures have been met:

$$p_B y = p_B z_B + (1 - \alpha)(m - p_A z_A - p_B z_B)$$

Note that the graph of $U(x, y)$ is a three-dimensional surface. An **indifference curve** is a contour line of the graph of the utility function defined by $U(x, y) = c$ for some constant c . At any point along an indifference curve, the individual is indifferent to possessing either x of good A or y of good B.

Marginal Utility

The **marginal utility** with respect to x (the quantity of some good X) is defined as:

$$g_X(x) = \frac{\delta U}{\delta x}$$

As just described, typically the marginal utility of a good will decrease as the quantity possessed of that good increases and will increase as the quantity of that good decreases. That is, typically as $x \rightarrow \infty$, $\frac{\delta U}{\delta x} \rightarrow 0$, and as $x \rightarrow 0^+$, $\frac{\delta U}{\delta x} \rightarrow \infty$. When using a Stone-Geary utility function, we will assume that consumption of a good necessary for subsistence is always equal to or greater than the amount required for subsistence.

The Marginal Rate of Substitution

The **marginal rate of substitution (MRS)** is the slope of the indifference curve defined by the utility function and represents the rate at which an individual will trade one good for another. The MRS is defined as:

$$h(x, y) = -\frac{\frac{\delta U}{\delta x}}{\frac{\delta U}{\delta y}}$$

The Budget Constraint

An individual's **budget constraint** reflects the maximum quantity x and y of goods A and B at prices p_A and p_B , respectively, that an individual can purchase given budget m :

$$p_A x + p_B y = m$$

where:

p_A is the price of good A,

p_B is the price of good B, and

m is income.

Utility Maximization Subject to the Budget Constraint

The **optimal bundle** $(x_{\text{optimal}}, y_{\text{optimal}})$ is defined as the amounts x and y of goods A and B, respectively, that maximize utility subject to the individual's budget constraint:

$$\max U(x, y) \text{ subject to } p_A x + p_B y = m$$

When considering a Stone-Geary utility function, the MRS is:

$$\frac{\frac{\delta U}{\delta x}}{\frac{\delta U}{\delta y}} = \frac{\alpha(y - z_B)}{(1 - \alpha)(x - z_A)}$$

The maximized utility subject to the budget constraint occurs when the slope of the MRS is equal to the slope of the budget constraint:

$$-\frac{\frac{\delta U}{\delta x}}{\frac{\delta U}{\delta y}} = -\frac{p_A}{p_B}$$

Demand Functions under Stone-Geary Utility

From the previous equation and the budget constraint, we can obtain the **demand functions** satisfying $\max U(x, y)$ subject to $p_A x + p_B y = m$:

$$x = z_A + \frac{\alpha}{p_A}(m - p_A z_A - p_B z_B)$$

and

$$y = z_B + \frac{1 - \alpha}{p_B}(m - p_A z_A - p_B z_B)$$

As mentioned earlier, we note that from these demand functions we can derive $p_A x$ and $p_B y$.

These demand functions therefore determine the individual's demand for goods A and B when goods A and B are at the optimal bundle $(x_{\text{optimal}}, y_{\text{optimal}})$ for constant m , p_A , p_B , z_A , and z_B .

The Effect of a Change in Price on the Price Elasticity of Demand

Monotonicity of the Price Elasticity of Demand

Suppose the opportunity cost of purchasing good B suddenly increases because the net benefit to an individual of purchasing good A increases from $f_0(A)$ to $f_1(A)$, such that $f_1(A) > f_0(A)$. We can model this by increasing the price of good B from p_{B0} prior to the net benefit increase of purchasing good A to p_{B1} after the net benefit increase of purchasing good A. Let the individual's utility function be a Stone-Geary utility function. Then, the individual's demand function for good B before the increase in net benefit is

$$y_0 = z_B + \frac{1 - \alpha}{p_{B0}}(m - p_A z_A - p_{B0} z_B)$$

and the individual's demand function for good B after the increase in net benefit is

$$y_1 = z_B + \frac{1 - \alpha}{p_{B1}}(m - p_A z_A - p_{B1} z_B)$$

So above, we have defined $p_{B0} < p_{B1}$, since the opportunity cost of purchasing good B, modeled as an increase in the price of good B, is increasing. Then, from the equations for y_0 and y_1 , we get:

$$\frac{dy_0}{dp_{B0}} = -\frac{1 - \alpha}{p_{B0}^2}(m - p_A z_A)$$

and

$$\frac{dy_1}{dp_{B1}} = -\frac{1 - \alpha}{p_{B1}^2}(m - p_A z_A)$$

The price elasticities of demand ϵ_{p_{B0}, y_0} and ϵ_{p_{B1}, y_1} for good B prior to and after the price increase, respectively, are:

$$\epsilon_{p_{B0}, y_0} = \frac{p_{B0}}{g(p_{B0})} \frac{dy_0}{dp_{B0}}$$

and

$$\epsilon_{p_{B1}, y_1} = \frac{p_{B1}}{g(p_{B1})} \frac{dy_1}{dp_{B1}}$$

where $y = g(p)$ is the demand for good B at price p .

Letting $\epsilon_{p_{B0}, y_0} < \epsilon_{p_{B1}, y_1}$, we get:

$$\frac{p_{B0}}{g(p_{B0})} \frac{dy_0}{dp_{B0}} < \frac{p_{B1}}{g(p_{B1})} \frac{dy_1}{dp_{B1}}$$

$$-\frac{p_{B0}}{z_B + \frac{1-\alpha}{p_{B0}}(m - p_A z_A - p_{B0} z_B)} \frac{1-\alpha}{p_{B0}^2} (m - p_A z_A) < -\frac{p_{B1}}{z_B + \frac{1-\alpha}{p_{B1}}(m - p_A z_A - p_{B1} z_B)} \frac{1-\alpha}{p_{B1}^2} (m - p_A z_A)$$

which simplifies to

$$p_{B0} < p_{B1}$$

Therefore, under a Stone-Geary utility function the price elasticity of demand is **monotonically increasing** (that is, $x \leq y \implies f(x) \leq f(y)$) and thus **an increase in the price of good B will increase the price elasticity of demand for good B**. This holds true for any good under a Stone-Geary utility function.

Limits of Price Elasticity of Demand

Next, note that $\epsilon_{p,y} \rightarrow -1$ as $p \rightarrow 0^+$ and $\epsilon_{p,y} \rightarrow 0$ as $p \rightarrow \infty$:

$$\lim_{p \rightarrow 0^+} \epsilon_{p,y} = -1$$

and

$$\lim_{p \rightarrow \infty} \epsilon_{p,y} = 0$$

Therefore, $\epsilon_{p,y} \in (-1, 0)$ and $|\epsilon_{p,y}| < 1$ (note that, because $p \neq 0$, $\epsilon_{p,y} \neq -1$ and $|\epsilon_{p,y}| \neq 1$).

Concerning $|\epsilon_{p,y}|$, note that $|\epsilon_{p,y}| \rightarrow 1$ as $p \rightarrow 0^+$ and $|\epsilon_{p,y}| \rightarrow 0$ as $p \rightarrow \infty$. That is:

$$\lim_{p \rightarrow 0^+} |\epsilon_{p,y}| = 1$$

and

$$\lim_{p \rightarrow \infty} |\epsilon_{p,y}| = 0$$

Therefore, under a Stone-Geary utility function **the price elasticity of demand for a good (with a nonzero price) will always be inelastic and will become more inelastic with increasing price.**

The Effect of a Change in Income on the Income Elasticity of Demand

Causes of Increasing Income Elasticity of Demand

Now consider the income elasticity of demand for good B:

$$\begin{aligned}
\epsilon_{m,q} &= \frac{m_0}{q_0} \frac{dq}{dm} \\
&= \frac{m_0}{h(m_0)} \frac{dq}{dm} \\
&= \frac{m_0(1-\alpha)}{p_B z_B + (1-\alpha)(m_0 - p_A z_A - p_B z_B)}
\end{aligned}$$

where $y = h(m)$ is the demand for good B at income m .

Let:

$$\epsilon_0 = \frac{m_0}{h(m_0)} \frac{dq}{dm} < \frac{m_1}{h(m_1)} \frac{dq}{dm} = \epsilon_1$$

where again $y = h(m)$ is the demand for good B at income m .

Then we get:

$$\frac{m_0(1-\alpha)}{p_B z_B + (1-\alpha)(m_0 - p_A z_A - p_B z_B)} < \frac{m_1(1-\alpha)}{p_B z_B + (1-\alpha)(m_1 - p_A z_A - p_B z_B)}$$

which simplifies to

$$(m_1 - m_0)(\alpha p_B z_B - (1-\alpha)p_A z_A) > 0$$

The result is then:

$$1 - \alpha < \frac{p_B z_B}{p_B z_B + p_A z_A}, \text{ if } m_0 < m_1$$

$$1 - \alpha > \frac{p_B z_B}{p_B z_B + p_A z_A}, \text{ if } m_0 > m_1$$

Therefore, under a Stone-Geary utility function **the income elasticity of demand for a good increases due to an *increase* in income when the proportion of subsistence**

expenditures (that is, income spent on subsistence levels of goods A and B) spent on subsistence levels of that good is greater than the proportion of leftover income spent on non-subsistence levels of that good.

Similarly, under a Stone-Geary utility function **the income elasticity of demand for a good increases due to a *decrease* in income** when the proportion of subsistence expenditures spent on subsistence levels of that good is less than the proportion of leftover income spent on non-subsistence levels of that good.

Causes of Decreasing Income Elasticity of Demand

Now let:

$$\epsilon_0 = \frac{m_0}{h(m_0)} \frac{dq}{dm} > \frac{m_1}{h(m_1)} \frac{dq}{dm} = \epsilon_1$$

which simplifies to

$$(m_1 - m_0)(\alpha p_B z_B - (1 - \alpha) p_A z_A) < 0$$

The result is then:

$$1 - \alpha > \frac{p_B z_B}{p_B z_B + p_A z_A}, \text{ if } m_0 < m_1$$

$$1 - \alpha < \frac{p_B z_B}{p_B z_B + p_A z_A}, \text{ if } m_0 > m_1$$

Therefore, under a Stone-Geary utility function **the income elasticity of demand for a good decreases due to an *increase* in income** when the proportion of subsistence expenditures spent on subsistence levels of that good is less than the proportion of leftover income spent on non-subsistence levels of that good.

Similarly, under a Stone-Geary utility function **the income elasticity of demand for**

a good decreases due to a *decrease* in income when the proportion of subsistence expenditures spent on subsistence levels of that good is greater than the proportion of leftover income spent on non-subsistence levels of that good.

Summary of Causes of Increasing and Decreasing Income Elasticity of Demand

These results are summarized in the following table, letting $s_B = \frac{p_B z_B}{p_B z_B + p_A z_A}$ (note that the choice of good is arbitrary—if good A were used, then we would look at the relationship between α and s_A):

Relationship of $1 - \alpha$ to s_B	Δm	$\Delta \epsilon_{m,q}$
$1 - \alpha < s_B$	$\Delta m > 0$	$\Delta \epsilon_{m,q} > 0$
$1 - \alpha > s_B$	$\Delta m < 0$	$\Delta \epsilon_{m,q} > 0$
$1 - \alpha > s_B$	$\Delta m > 0$	$\Delta \epsilon_{m,q} < 0$
$1 - \alpha < s_B$	$\Delta m < 0$	$\Delta \epsilon_{m,q} < 0$

Limits of Income Elasticity of Demand

Next, note that $\epsilon_{m,q} \rightarrow 0$ as $m \rightarrow 0^+$ and $\epsilon_{m,q} \rightarrow 1$ as $m \rightarrow \infty$. That is:

$$\lim_{m \rightarrow 0^+} \epsilon_{m,q} = 0$$

and

$$\lim_{m \rightarrow \infty} \epsilon_{m,q} = 1$$

The same holds true for $|\epsilon_{m,q}|$:

$$\lim_{m \rightarrow 0^+} |\epsilon_{m,q}| = 0$$

and

$$\lim_{m \rightarrow \infty} |\epsilon_{m,q}| = 1$$

Therefore, under a Stone-Geary utility function **the income elasticity of demand for a good is always inelastic and may become either more elastic or more inelastic depending on the proportion of subsistence expenditures spent on subsistence levels of that good and the proportion of leftover income spent on non-subsistence levels of that good. However, as income becomes sufficiently small, the income elasticity of demand becomes more inelastic, and as income becomes sufficiently large, the income elasticity of demand becomes more elastic.**

Biological Needs

Biological needs are those needs required for the survival of an organism. Increasing access to these biological needs increases an organism's probability of survival by decreasing its risk of disability and premature mortality. Naturally, by increasing the probability of an organism's survival, these needs also increase the period of time during which an organism can reproduce, thus increasing the reproductive fitness of the organism. Therefore, the motivation to acquire these needs is naturally selected for.

This paper will focus on the biological needs of humans. Those biological needs required for basic physiological processes include:

- Clean water
- Nutritious food
- Clean air
- Adequate sleep
- Physical and psychological safety

An individual requires sufficient water, food, air, sleep, and physical and psychological safety to survive. By increasing the individual's health, each of these increases reproductive fitness as well. Note that the terms "clean," "nutritious," and "adequate" above are not optional.

Contaminants in water, food, or air harm the individual's health and thus increase the risk of disability and premature mortality, decreasing the probability of survival and thus decreasing the probability of reproducing. Inadequate sleep—in terms of both quantity (sufficient hours slept) and quality (sleep occurring in a safe, quiet, dark, and temperature-controlled environment)—damages the individual's health, decreases productivity, and increases the risk of harm from external sources, all of which increase the risk of disability and premature mortality and therefore threaten survival and reproduction. Obviously, increased risk of physical or psychological harm also threatens the individual's survival and reproduction as well.

Biological needs do not end there, however, as several other requirements exist for survival, including:

- Education
- Medical care
- Safe, spacious housing
- Reproduction

Each of these are essential for survival. Education is required to ensure that an individual can work to acquire other biological needs such as food and housing for himself or herself, his or her partner, and their children. Education is also required to avoid danger, attract a partner, and raise children. Medical care is obviously required to prevent disability and premature mortality. Safe, spacious housing is required for numerous reasons, including for the individual to sleep safely (obtaining adequate sleep in the absence of physical or psychological harm from threats in the environment) and raise children.

As a result of natural selection, a healthy individual will typically be motivated to reproduce to pass on his or her genes to offspring. Of course, some individuals do not reproduce and this does not appear to impact their survival. However, because reproduction is essential for life it is a biological need on a population level.

The opportunity cost of forgoing a biological need for an extended period of time is extremely large because the result of forgoing such a need significantly increases the risk of disability and

premature mortality, thus significantly decreasing the probability of survival and reproduction. Except for the opportunity cost of forgoing other biological needs, there are very few actions for which the opportunity cost of forgoing such an action for an extended period of time would rival the large cost of forgoing a biological need for that same period. Thus, from the previous discussion, if we assume that any given individual has a Stone-Geary utility function then it must be the case that **the price elasticity of demand for biological needs is extremely inelastic**. As expected, existing research supports the inelasticity of biological needs such as food,¹ water,² and housing³.

Exchange of Labor for Biological Needs

In an employee-employer relationship, an employee and employer enter into an agreement in which the employee agrees to exchange his or her labor for a wage. As part of this transaction, the employee accepts not only the benefit that is the wage but also, if applicable, the cost of poor working conditions or long working hours. In addition, the employee forgoes other actions and thus must take into account the opportunity cost of his or her employment. Most employees cannot afford to meet all of their biological needs without work, while employers typically have more resources to obtain these needs due to the capital they possess. This imbalance gives employers a major advantage over employees in the negotiation of a wage, hours worked, and working conditions.

Suppose that an employee has a Stone-Geary utility function. As the previous discussion shows, **since demand for biological needs is highly inelastic a worker will pay a higher price in order to obtain them**. Therefore, the opportunity cost of forgoing work—which provides the means (money) to acquire biological needs—is extremely high. This cost may be reduced if the worker is able to acquire some of his or her biological needs by other methods, such as financial support from family or the government.

From the previous discussion, **since the price elasticity of demand for biological needs is extremely inelastic, decreasing resources (money) to acquire these needs without work (including increasing price of biological needs in the market and increasing taxation without an offsetting benefit) will result in the employee paying**

¹Andreyeva et al., 2010

²Bruno & Jessoe, 2021

³Albouy et al., 2016

increasing prices for biological needs. In an employee-employer relationship, this higher price is reflected in an employee's acceptance of a lower wage, longer or worse working hours, and worse working conditions than he or she would otherwise accept. Once again, the fewer resources an employee has available to him or her to fulfill his or her biological needs without work, the more the employee will offer up to the employer during the negotiation over wage, hours worked, and working conditions.

A similar effect is seen from the previous discussion of income elasticity of demand. **If the proportion of subsistence income an employee spends on biological needs is greater than the proportion of leftover income he or she spends on those needs, then as the employee's income decreases, the price of biological needs in the market increases, or taxes increase without an offsetting benefit to the employee, the income elasticity of demand for biological needs becomes more inelastic. Furthermore, as an employee's income becomes sufficiently small, the price of biological needs in the market becomes sufficiently large, or the difference between taxes and the benefit to the employee becomes sufficiently large, the income elasticity of demand for the employee's biological needs becomes more inelastic.**

The **value an employer extracts from an employee** during negotiations due to the employee's inability to acquire his or her biological needs without work is equivalent to **the difference between the area under the more inelastic demand curve that results from scarcity and the area under the more elastic demand curve that would result if the employee had the resources necessary to fulfill all of his or her biological needs without needing to work.** Let $y_0 = f_0(p)$ be the more elastic demand curve due to sufficient resources and $y_1 = f_1(p)$ be the more inelastic demand curve due to scarcity. Then, the value extracted by an employer from an employee during the negotiation of a wage, hours worked, and working conditions is:

$$v_{extracted} = \int_0^{p_E} f_1(p) - f_0(p) dp$$

where:

p_E is the equilibrium price (the price at which the supply and demand curves intersect), and p is the price.

When $v_{extracted} > 0$, the employer has extracted value from the employee through the employee's need to accept a lower wage, worse working hours, or worse working conditions than he or she otherwise would in order to survive and reproduce due to his or her inability to obtain biological needs without work.

The mechanisms by which a lower wage may be paid include lack of proper adjustment for cost of living, increased working hours, or underemployment, a situation in which an employee qualified for a higher paying job is forced to work a lower paying job that does not require the employee's qualifications. Worse working hours can include working during holidays or overnight, working long shifts, working a fixed schedule of disparate shifts (such as working mornings some days and evenings other days), working on-call or without a fixed schedule, and working mandatory overtime. Poor working conditions can include conditions that are hazardous to physical or psychological health, such as conditions that are cramped, poorly ventilated, or lack sunlight. Working conditions that specifically harm psychological health also include those in which the employee is not treated with the dignity deserved by (and required for the psychological health of) a human.

Labor Exploitation

In a capitalist system in which shareholders own corporations that hire employees, shareholders (the employers) extract value produced by the company's employees without themselves having to work for this value. If there are not external regulatory mechanisms in place to ensure that employees are guaranteed their basic biological needs, shareholders, in the absence of internal regulatory mechanisms such as empathy or morality, will tend to extract as much value as possible from their employees' labor. This can create a vicious cycle in which shareholders extract increasing amounts of value from employees, decreasing the ability of employees to increase their own wealth in order to escape the cycle through acquisition of more of their biological needs and thus allowing them to negotiate better wages, working hours, and working conditions.

Meanwhile, shareholders—who are disproportionately older, since they have had more time to accumulate wealth than younger people—increase the risk of harm and premature mortality to employees—who will be disproportionately younger—by extracting increasing amounts of

value from them. As more value is extracted from the young by the old, the youth experience decreasing health and increasing premature mortality. Signs of decreasing youth health include increasing burden of physical disease and mental illness, an increasing rate of substance abuse, an increasing rate of loneliness (due to a decreasing rate of socialization), and an increasing rate of suicide within this population. Another sign of decreasing youth health is decreasing fertility, primarily attributable to decreasing access to biological needs necessary to raise children, especially housing (driven by artificially reduced supply and poor transportation) and childcare (two parents are now typically required to pursue higher education and work full-time to afford modest rental housing, whereas this was not the case only several decades ago). Collectively, these processes may be described as the “cannibalization of the young.”

This process is playing out in developed countries on a global scale. The wealth of those older than 60 continues to accelerate at a rapid pace while the wealth of those younger than 40 continues to remain stagnant. This is despite the fact that young people today are more educated—especially in highly economically productive fields such as science, technology, engineering, mathematics, and medicine (STEMM)—and more productive than older people were when they were young employees themselves. As such, the real median wage of young males would be expected to increase. Instead, largely due to the siphoning of value by the older population from the young, it has actually *decreased* (young males are used for comparison, since female representation in higher education and participation in the full-time labor force has drastically increased over the past several decades). When compared to output, this decline is even more striking.

Let output be defined as:

$$\omega = \phi \times t$$

where:

ϕ is the total factor productivity, and

t is the time worked.

Then, the real median income (RMI) per unit output indicates real compensation for some unit of output, allowing for comparison across years while taking into account inflation and

fluctuations in productivity and the number of hours worked in a year. RMI per unit output is defined as:

$$\frac{\bar{w}_{\text{real}}}{\omega}$$

where $\bar{w}_{\text{real}}$ is the real median income.

As shown by Figure 1, RMI per unit output for males aged 25 to 34 rose from 1960 to 1973 and remained elevated until 1979 before plummeting. Although small recoveries were made in the 1990s and 2010s, RMI per unit output for males is drastically below even its value in 1960 and continues to decline. This is despite the fact that productivity has drastically increased and hours worked have fallen only slightly, yielding a large increase in output by workers.⁴ The real median income has not kept up, thus much of the value of the increased output must be going elsewhere—specifically, to shareholders.

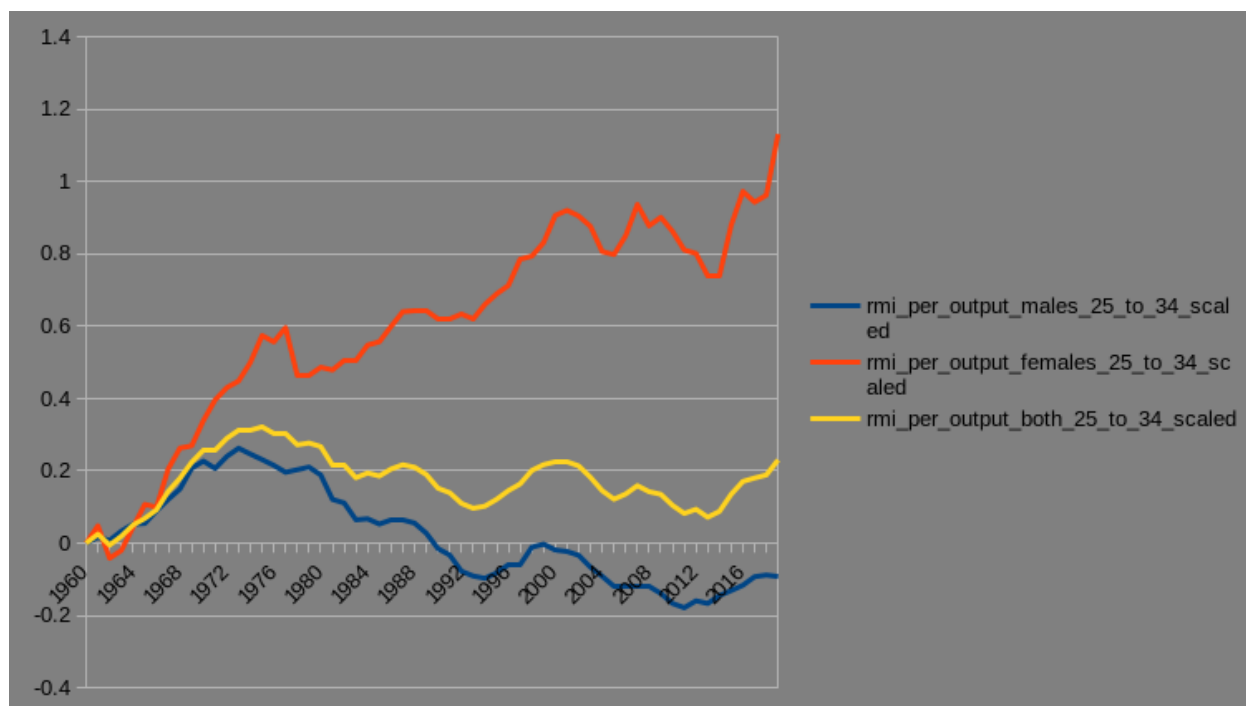


Figure 1: RMI per Unit Output for Males Ages 25 to 34, Scaled. RMI per unit output has decreased for males aged 25 to 34 since the 1970s, despite increased total factor productivity and only slightly decreased working hours. Overall RMI per unit output has

⁴University of Groningen, n.d.-a; University of Groningen, n.d.-b

slightly increased due to increased female participation in the full-time labor force and increased female attainment of higher education.

Moreover, RMI per unit output demonstrates the flaws of overreliance on RMI as an indicator of economic prosperity. As shown in Figure 1, RMI may change when a large demographic changes the way in which it engages with the labor market. In this example, RMI increased as women increasingly transitioned from part-time work or no work to full-time work and from jobs requiring a high school degree or less to jobs requiring more advanced education.⁵ Therefore, over the past several decades RMI has largely captured increasing engagement of women with the labor force rather than broadly increasing economic prosperity for most workers. In fact, as the *declining* RMI per output for young males shows, the economic benefits reaped over the past several decades have largely been captured by corporate shareholders via labor exploitation. This has—in combination with drastically increasing housing costs (due to artificially reduced supply), increasing child care costs, and high taxes on the youth to subsidize government spending that disproportionately benefits the older population—drastically reduced fertility in the United States.⁶ The same mechanism has drastically reduced the fertility rate in most “developed” nations as well as many “developing” nations.⁷ Notably, in the United States—where labor exploitation is the most extreme among “developed” nations—this crisis has been accompanied by markedly increased youth morbidity and mortality.⁸ Humanity is “cannibalizing” its young—both current and future—for the immediate benefit of the present-day old.

Solutions to these crises exist and are highly cost-effective. The most obvious and most vital solution is to mitigate labor exploitation by dramatically expanding workers’ access to biological needs without work and curtailing the means by which exploitation is effectuated and intensified. This can be achieved through requiring all workers along a company’s supply chain to be paid a minimum hourly effective wage greater than or equal to the cost of living; rapidly increasing the supply of dense, moderately-sized, and rail-proximal housing and increasing rail-oriented transit development;⁹ decreasing the cost of education for STEM and similar programs; increasing the quantity of nuclear power plants and decreasing fossil fuel

⁵U.S. Bureau of Labor Statistics, 2022

⁶Lino et al., 2017; Madgavkar et al., 2025; United Nations, 2025

⁷United Nations, 2025

⁸Forrest et al., 2025

⁹Burchell et al., 2002; Miner et al., 2024; Schröder et al., 2023

use;¹⁰ establishing a universal healthcare system;¹¹; incentivizing decreased labor exploitation by preferentially investing in corporations that pay workers higher effective wages; and severely reducing taxes on young people and increasingly implementing taxation based on use (for example, taxation to pay for roads would be a fee based on miles driven and vehicle weight) and externalities (for example, taxation to pay for healthcare costs resulting from the externalities of automobile travel would be a fee based on miles driven, vehicle weight, vehicle fuel efficiency, and population near the roads traveled).

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¹⁰International Energy Agency et al., 2020; Kharecha & Hansen, 2013; Markandya & Wilkinson, 2007

¹¹Bichay, 2020; Molander, 2025; Organisation for Economic Co-operation and Development, 2023; Wager & Cox, 2024

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